

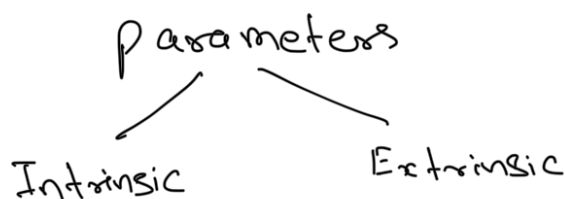
Camera Calibration

First question which CV starts with:

Is there any correspondence between
distances in real-world and distances
in image plane
[There has to be a correspondence
near objects in real world should
be near in image plane
far objects should be far in
image plane

In some applications (eg autonomous vehicle)
we need exact relation between
distance in real world and distances
in image plane.

Camera Calibration: It is the process of
getting 'parameters' which are useful
in mapping of points in real world
Coordinate system with corresponding
points in image Coordinate system.



Parameters

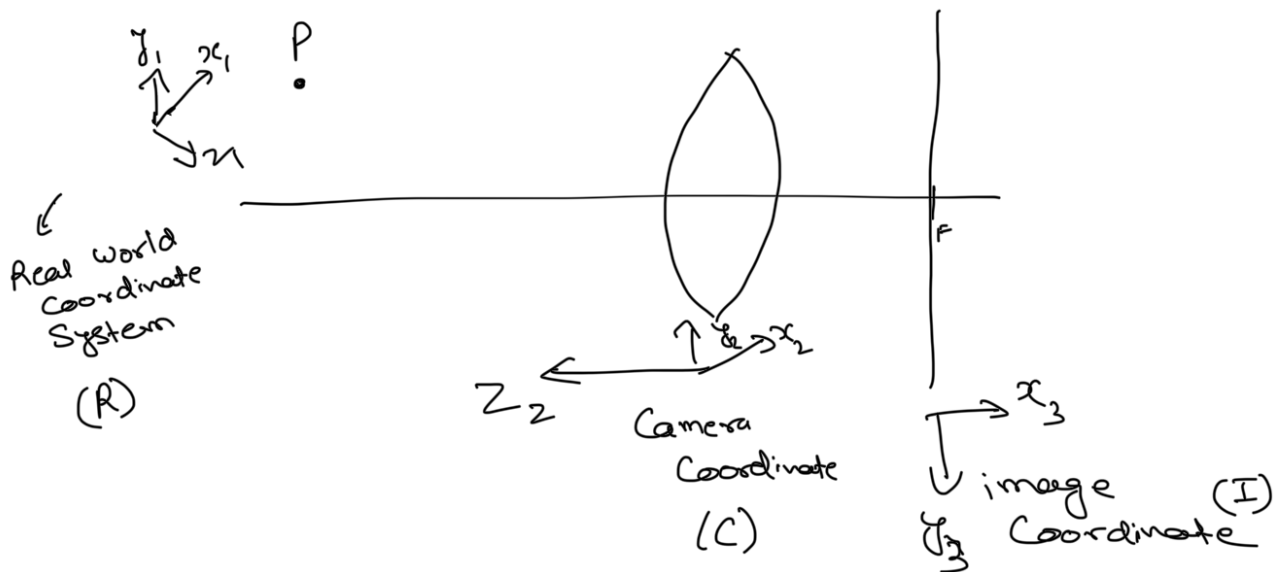
Used to transform
Coordinates from
Camera Coordinate
to image
Coordinate System
eg: focal length

Parameters

Used to transform
Coordinates from
real world Coordinate
System to image
Coordinate System

eg: $\underbrace{t_x, t_y, t_z}_{\text{translation}}, \underbrace{r_x, r_y, r_z}_{\text{rotation}}$

Following are the three coordinate system
(in image acquisition process)



When we acquire an image, the objects in
'R' coordinate system are mapped to
points in the 2D coordinate system (I)
Image plane is 2D

We are interested in a method of
matrix transformation by which

we are able to move points
from one coordinate system
to another coordinate system

e.g. R to C

$$\downarrow$$
$$\begin{bmatrix} x_2 \\ y_2 \\ z_2 \end{bmatrix} = K_{ext} \begin{bmatrix} x_1 \\ y_1 \\ z_1 \end{bmatrix}$$

$$C \text{ to } I \quad \begin{bmatrix} x_3 \\ y_3 \\ z_3 \end{bmatrix} = K_{int} \begin{bmatrix} x_2 \\ y_2 \\ z_2 \end{bmatrix}$$

To understand form of K_{int} and K_{ext}

↓
let's first understand effect
of matrix transformation
on coordinates and images.

Image Transformation as Matrix Operation (Extrinsic Parameters)

1) Scaling

$$\text{old coordinate} \Rightarrow \begin{bmatrix} x \\ y \end{bmatrix} \Rightarrow \text{new coordinates} \Rightarrow \begin{bmatrix} sx \\ sy \end{bmatrix}$$

I' $\Rightarrow$ new image

$I \Rightarrow$ old image

$$I'(s_x, s_y) = \underline{I(x, y)}$$

i.e. value at pixel location (s_x, s_y)
in new image is
taken from value at
 (x, y) in old image.

Also known as scaling using forward mapping

As coordinates (x', y')
of new image are
represented in terms of coordinates
of old image (x, y)

$$x' = s_x \quad , \quad y' = s_y$$

If is desired, that these operations should
be available in matrix format as
it allows for efficient implementation.

In matrix format

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} s & 0 \\ 0 & s \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$$

Representation of scaling operation in matrix
format

2) Rotation:

Represent coordinates in polar form
old $\Rightarrow x = r \cos \phi$

$$y = r \sin \phi$$

Rotate the point by an angle Θ (in anticlockwise)

$$x' = r \cos(\phi + \Theta)$$

$$y' = r \sin(\phi + \Theta)$$

$$x' = r \cos \phi \cos \Theta - r \sin \phi \sin \Theta$$

$$y' = r \sin \phi \cos \Theta + r \cos \phi \sin \Theta$$

$$\begin{cases} x' = x \cos \Theta - y \sin \Theta \\ y' = y \cos \Theta + x \sin \Theta \end{cases}$$

↓
In matrix form

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \underbrace{\begin{bmatrix} \cos \Theta & -\sin \Theta \\ \sin \Theta & \cos \Theta \end{bmatrix}}_{\text{Rotation Matrix}} \begin{bmatrix} x \\ y \end{bmatrix}$$

3)

Translation

$$x' = x + t_x$$

$$y' = y + t_y$$

$$\underbrace{\begin{bmatrix} x' \\ y' \end{bmatrix}}_{\text{new coordinate}}$$

$$\underbrace{\begin{bmatrix} x \\ y \end{bmatrix}}_{\text{old coordinate}}$$

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} x \\ y \end{bmatrix} + \begin{bmatrix} t_x \\ t_y \end{bmatrix}$$

i.e Translation in normal coordinate

system is just vector addition

(not a matrix operation)
 Not good for pipelines as we want to have a matrix for each transformation

⇒ Represent each point (x, y) as $(x, y, 1)$
 known as homogeneous coordinate system

Translation in homogeneous coordinate

$$\begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & t_x \\ 0 & 1 & t_y \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$

Thus Translation is a matrix operation in homogeneous coordinate system.

Homogeneous Coordinate ⇒ Add an additional component (value = 1) to the original coordinate system

Scaling ⇒

$$\begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} = \begin{bmatrix} s & 0 & 0 \\ 0 & s & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$

Rotation ⇒

$$\begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} = \begin{bmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$



=> Want to do combination of Transformation

multiply the matrix

$$\text{e.g. } \begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} = ST \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$

Translation followed by Scaling

Note: Order of operation is important as matrix multiplication is not commutative.

In all applications, extrinsic matrix is evaluated as a combination of transformations such as scaling, rotation and translation. //